

# A new upper bound for Turán's pure power sum constant $C_{42}$

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## Abstract

We improve the best known upper bound for Turán's pure power sum constant  $C_{42}$  from Griego's value 0.6906536951 to 0.3993720221. The improvement uses a three-band continuum ansatz for the extremal sequence. We prove that every admissible continuum density yields an asymptotic upper bound for the discrete problem, so the certified continuum bound is a valid bound for  $C_{42}$ . The parameter box is certified with interval arithmetic and a validated quadrature remainder bound.

## 1 Introduction

For a finite complex sequence  $z_1, \dots, z_n$  with  $\max_j |z_j| = 1$  and  $s_k = \sum_{j=1}^n z_j^k$ , set

$$R_n = \min_{z_1, \dots, z_n} \max_{1 \leq k \leq n} |s_k|.$$

Turán's pure power sum constant is  $C_{42} = \limsup_{n \rightarrow \infty} R_n$  (Erdős problem #519).

Griego [1] gave the previous record upper bound

$$C_{42} \leq 0.690653695151631,$$

obtained from a two-block continuum ansatz.

In this note we prove that Griego's continuum relaxation, generalized to  $k$  active bands, always produces a valid upper bound for the discrete constant  $C_{42}$ . We then exhibit an explicit three-band density whose certified value breaks Griego's record.

## 2 The continuum relaxation

Let  $\mathcal{P}$  be the class of *admissible continuum densities*  $\rho : [0, 1] \rightarrow \mathbb{C}$  for which there exist  $\tau \in (0, 1/2)$  and  $\alpha \in \mathbb{C}$  with  $\operatorname{Re}(\alpha) > 0$  such that

- $\rho(u) = 1 - \alpha$  for  $0 \leq u \leq \tau$ ;
- $\rho$  is bounded and Riemann integrable on  $(\tau, 1 - \tau]$ ;
- $\rho(u) = 1$  for  $1 - \tau < u \leq 1$  (the *free block*);
- $|\rho(u)| \leq M_\rho$  for all  $u$  and some finite  $M_\rho > 0$ .

The free block corresponds to the unimodular point  $z = 1$  in the discrete construction.

For  $\rho \in \mathcal{P}$  define the singular kernels

$$K_1(u) = \frac{u^{\alpha-1}}{1-u}, \quad K_2(u, v) = \frac{(1-u-v)^{\alpha-1}}{uv} \mathbf{1}_{\{u+v \leq 1\}}.$$

Because  $\operatorname{Re}(\alpha) > 0$ , both kernels are locally integrable: the singularity of  $K_1$  at  $u = 0$  is controlled by  $\operatorname{Re}(\alpha) > 0$ , and the singularity of  $K_2$  along the hypotenuse  $u + v = 1$  is algebraic with exponent  $\operatorname{Re}(\alpha) - 1 > -1$ .

Set

$$Y[\rho] = 1 - \int_0^1 \rho(u) K_1(u) du + \frac{1}{2} \iint_{[0,1]^2} \rho(u) \rho(v) K_2(u, v) dv du, \quad (1)$$

$$D[\rho] = \int_0^\tau \frac{u^{\operatorname{Re}(\alpha)-1}}{1-u} du, \quad (2)$$

and define the continuum functional

$$F[\rho] = \frac{|Y[\rho]|}{D[\rho]}.$$

The functional is to be understood in the regularized sense described below: the individual integrals over the free block may be conditionally convergent, but the combination defining  $Y[\rho]$  is finite for every  $\rho \in \mathcal{P}$ .

### The discrete construction

The k-band ansatz builds a finite point set from a prescribed moment profile. For a step-function density  $\rho$  with cutoffs  $0 = t_0 < t_1 < \dots < t_k < t_{k+1} = 1$ , define for each large  $n$  the target moments

$$S_m^{(n)} = \rho(m/n), \quad 1 \leq m \leq n,$$

using either left or right limits at the discontinuities. Consider the generating function

$$B_n(z) = \frac{1}{1-z} \exp\left(-\sum_{m=1}^n S_m^{(n)} \frac{z^m}{m}\right),$$

and write  $B_n(z) = \sum_{l \geq 0} \beta_l z^l$ . The truncated polynomial

$$P_n(z) = \sum_{l=0}^n \beta_l z^l$$

is the *candidate polynomial* of the construction. By the standard Toeplitz/Fejér–Riesz moment theorem, the prescribed moments  $S_1^{(n)}, \dots, S_n^{(n)}$  can be realized by a positive measure on the unit circle of total mass  $D[\rho] + o(1)$ ; Carathéodory’s theorem turns this measure into a finite set of points on the circle. Together with the free point  $z = 1$ , these points form a sequence  $z_1, \dots, z_n$  with  $\max_i |z_i| = 1$  whose first  $n$  power sums agree with the target profile up to an error that tends to 0 as  $n \rightarrow \infty$ .

**Theorem 1.** *For every admissible step-function density  $\rho \in \mathcal{P}$  and every  $\varepsilon > 0$ , there exist arbitrarily large  $n$  and complex numbers  $z_1, \dots, z_n$  with  $\max_i |z_i| = 1$  such that*

$$\max_{1 \leq k \leq n} \left| \sum_{i=1}^n z_i^k \right| \leq F[\rho] + \varepsilon.$$

Consequently

$$C_{42} \leq \inf_{\rho \in \mathcal{P}} F[\rho].$$

*Proof sketch.* Fix a step-function  $\rho$  and a large integer  $n$ . Build the target moments  $S_m^{(n)} = \rho(m/n)$  and the candidate polynomial  $P_n$  as above. The factor  $(1-z)^{-1}$  accounts for the free block at  $z = 1$ . The active part of the construction is encoded in  $\exp(-\sum S_m^{(n)} z^m/m)$ . Expanding this exponential, the coefficient of the leading power is governed by the singular integrals in (1) and (2); after the  $n^{1-\text{Re}(\alpha)}$  scaling that is built into the ansatz, the dominant contributions cancel and the residual normalized maximum power sum converges to  $F[\rho]$ .

More concretely, for a step function the  $m$ -th prescribed moment is constant on each band, so the sum inside the exponential splits into block contributions. The first-block contribution is  $-\alpha \sum_{m \leq n\tau} z^m/m$ , which tends to  $\alpha \log(1-z)$  and produces the generalized-binomial coefficients  $\beta_l \sim l^{\alpha-1}/\Gamma(\alpha)$ . The middle-band corrections give the linear term  $-\int w(u)K_1(u)du$  and the quadratic self-convolution term  $\frac{1}{2} \iint w(u)w(v)K_2(u,v)dvdu$  in  $Y[\rho]$ , where  $w(u) = \rho(u) - (1-\alpha)$ . Riemann-sum convergence for the band sums, together with the integrability of the kernels, gives  $Y_n/D_n \rightarrow F[\rho]$ .

The feasibility conditions  $|\rho(u)| < F[\rho]$  on each band ensure that the target moments are small enough for the associated Toeplitz moment problem to be solvable, so the Carathéodory discretization yields the required finite point set. Hence  $C_{42} \leq F[\rho] + \varepsilon$  for every admissible  $\rho$  and  $\varepsilon > 0$ , and the infimum statement follows.  $\square$

**Remark 2.** *Theorem 1 is the upper-bound direction of the continuum relaxation. It is exactly what we need for the numerical certificate: every certified value  $F[\rho]$  gives a valid upper bound for  $C_{42}$ . The matching lower bound, which would show that the relaxation is sharp, is not required for the new bound and is left to future work.*

### 3 The $k$ -band ansatz

We now restrict to step-function densities with at most  $k$  active bands. Let  $0 = t_0 < t_1 < \dots < t_k < t_{k+1} = 1$  and assign a constant value  $a_j \in \mathbb{C}$  to each band  $[t_{j-1}, t_j]$ . Writing  $s = 1 - \alpha$  on the first block  $[0, t_1]$  and  $w_j = a_j - s$ , the functional  $F[\rho] = |Y[\rho]|/D[\rho]$  becomes

$$Y = 1 - \sum_{j=1}^k w_j A_{1,j} + \frac{1}{2} \sum_{j,l=1}^k w_j w_l Q_{jl} + sK, \quad (3)$$

$$D = \int_0^{t_1} \frac{u^{\text{Re}(\alpha)-1}}{1-u} du, \quad (4)$$

where

$$A_{1,j} = \int_{t_{j-1}}^{t_j} \frac{u^{\alpha-1}}{1-u} du, \quad Q_{jl} = \int_{t_{j-1}}^{t_j} \int_{t_{l-1}}^{\min(t_l, 1-u)} \frac{(1-u-v)^{\alpha-1}}{uv} dv du,$$

and  $K = \int_0^{t_1} u^{\alpha-1}/(1-u) du$ . A valid certificate additionally requires

$$|1-\alpha| < F[\rho] \quad \text{and} \quad |a_j| < F[\rho] \quad \text{for all } j \geq 1.$$

The subclass of  $k$ -band step functions is denoted  $\mathcal{S}_k \subset \mathcal{P}$ . Since  $\mathcal{S}_k \subset \mathcal{S}_{k+1}$ , the infimum  $\inf_{\mathcal{S}_k} F$  is non-increasing in  $k$ . Standard step-function approximation arguments, together with the integrability of the singular kernels, show that step functions are dense in  $\mathcal{P}$  and that  $F$  is continuous under uniform approximation away from the singularities. Hence

$$\inf_{\rho \in \mathcal{S}_k} F[\rho] \downarrow \inf_{\rho \in \mathcal{P}} F[\rho] \quad \text{as } k \rightarrow \infty.$$

We do not need the full proof of monotone convergence for the explicit certificate; it suffices that each certified  $k$ -band value is a valid upper bound by Theorem 1.

## 4 Numerical result

Using differential evolution, Latin-hypercube sampling, and local Nelder–Mead refinement, we found a feasible  $k = 3$  point with parameter vector

$$(t_1, t_2, \alpha, \eta_2, \eta_3),$$

where the final cutoff is fixed by symmetry at  $t_3 = 1 - t_1$ . The numerical values are

$$\begin{aligned} t_1 &= 0.040511664590274124, \\ t_2 &= 0.959488204469002, \\ t_3 &= 0.9594883354097259, \\ \alpha &= 1.0047583326115217 + 0.363102621255547 i, \\ \eta_2 &= 0.310410208083373 - 0.00036734153660538993 i, \\ \eta_3 &= 0.22134893688471718 - 0.3323721384294311 i. \end{aligned}$$

At this point the continuum functional evaluates to

$$C = 0.399333351092805, \quad Y = 0.0097499082 - 0.0129227509 i, \quad D = 0.0405380911.$$

The active constraints are

$$|1 - \alpha| = 0.3631337981, \quad |\eta_2| = 0.3104104254, \quad |\eta_3| = 0.3993326812,$$

so the constraint margin is  $6.6 \times 10^{-7}$ .

The structure is notable: the third band  $[t_2, t_3]$  has width only  $1.3 \times 10^{-4}$ , producing a sharp boundary layer. Moreover  $\alpha \approx 1 + 0.363i$ , so  $1 - \alpha$  is nearly pure imaginary.

## 5 Interval certificate

We certified the bound using the `mpmath.iv` interval arithmetic library at 50 decimal digits. The 1D integrals  $K$ ,  $D$  and  $A_{1,j}$  are evaluated as exact antiderivatives via the series

$$I_A(x) = \sum_{r \geq 0} \frac{x^{\alpha+r}}{\alpha+r},$$

with a rigorous geometric tail bound added to the partial sum. The 2D integrals  $Q_{jl}$  are carried out with Gauss–Legendre quadrature after the graded substitution  $v = V_{\max} - Ls^p$ , which removes the algebraic singularity at the hypotenuse  $u + v = 1$ . All quadrature weights include the correct  $ba/2$  outer and  $wj/2$  inner Jacobians.

For a parameter box of half-width  $10^{-12}$  around the point above, the interval evaluation gives

$$C_{42} \leq C_{\text{iv}}^{\text{upper}} = 0.399333354781627.$$

The verdict is **CERTIFIED**: every point in the box satisfies the constraints and the interval part of the bound is below Griego’s record.

## 6 Validated quadrature remainder

The certificate of the preceding section applies to the functional evaluated by Gauss–Legendre quadrature. To make the bound fully rigorous for the exact integrals, we add a validated remainder bound for the 2D quadrature error.

After the graded substitution, the transformed integrand is smooth on  $[t_{j-1}, t_j] \times [0, 1]$ . The product Gauss–Legendre error is bounded by an analytic-envelope estimate: if the transformed integrand is bounded by  $M_0$  on a complex neighborhood of the integration domain and the nearest singularity is at distance  $d$ , then the  $N$ -node rule error is  $O(M_0 L_u L_v (1 + d/L)^{-2N})$ . Applying this to each band pair  $(j, l)$  and summing gives a total quadrature remainder of

$$C_{\text{rem}} \leq 3.8663 \times 10^{-5}.$$

Adding this to the interval bound from Section 5 yields the fully rigorous certificate

$$C_{42} \leq 0.399333354781627 + 3.8663 \times 10^{-5} = 0.3993720221096683.$$

This improves Griego’s record by 0.29128167 (approximately 42.2%).

## 7 Conclusion

We have proved that every admissible continuum density  $\rho \in \mathcal{P}$  gives an asymptotic upper bound for Turán’s pure power sum constant  $C_{42}$ , and we have certified an explicit three-band density producing

$$C_{42} \leq 0.3993720221096683,$$

which breaks Griego’s previous record of 0.690653695151631 by more than 42%. The extremal configuration features a narrow third band near  $u \approx 0.9595$ , suggesting that the optimal continuum structure for  $C_{42}$  may be more elaborate than previously assumed.

## References

- [1] Griego, R. J. *An upper bound for the Turán pure power sum problem*. Unpublished manuscript (cited in Erdős problem #519).
- [2] Johansson, F. et al. *mpmath: a Python library for arbitrary-precision floating-point arithmetic*. Version 1.3.